

Process Characterization Plan

**A-Mab Drug Substance — Small-Virus Retentive Filtration (Step 9) —
PCP-009**

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

Table of contents

Approvals	3
Abbreviations	3
1 Purpose and scope	5
2 Objectives	6
3 Related documents	7
4 Prior knowledge and quality risk basis	9
4.1 Unit-operation description and prior knowledge	9
4.2 Quality attributes in scope	10
4.3 Risk-based prioritization of parameters	10
5 Materials and methods	12
5.1 Scale-down model and its qualification	12
5.2 Operation	12
5.3 Analytical methods	13
5.4 Sampling plan	14
5.5 Statistical methods	14
6 Study design	16
6.1 Factors, ranges and study type	16
6.2 Screening design	17
6.3 Response-surface design	17
6.4 Univariate assessment	17
7 Acceptance and decision criteria	19
8 Proven acceptable ranges (planned analysis)	21
9 Data management and integrity	23
10 Roles and responsibilities	24
11 Deliverables and schedule	25
12 Risks and assumptions	26
13 Approvals	27
14 Appendices A and B — Planned design matrices	28
14.1 Appendix A — Planned screening design matrix	28

14.2Appendix B — Planned response-surface design matrix	29
---	----

References 30

Field	Value
Document ID	PCP-009
Document class	Process Characterization Plan
Title	Process Characterization Plan: Small-Virus Retentive Filtration (Step 9)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

SYNTHETIC DOCUMENT — NOT A REAL RECORD. Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ALCOA+, attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available; AMV, analytical method validation; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; ICH, International

Council for Harmonisation; LOQ, limit of quantitation; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PRESS, predicted residual error sum of squares; RPN, risk priority number; RSM, response surface methodology; SOP, standard operating procedure; TCID₅₀, median tissue culture infective dose; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

Small virus retentive filtration is the dedicated virus removal step of the A-Mab drug substance process, and this plan defines the studies that will characterize it. The step removes virus by size, inside the structure of the membrane, and because it works on a principle that has nothing in common with the low-pH hold at Step 6 or with the anion exchange step at Step 8, its clearance is claimed as a separate module of the viral safety argument (International Council for Harmonisation 2023a). This plan states what will be studied, how the studies will be executed and analysed, and the criteria against which the results will be judged. It is written before any data exist and contains no findings.

The scope is the 2 operating parameters of the step, Filtration volume (load) and Filtration pressure, studied together over the characterization ranges given in Table 6.1. Both will be varied in a multivariate design on a qualified scale-down spiking model, using the two model viruses that bound the retention challenge at this step and are reported throughout this package: minute virus of mice (MVM) as the small non-enveloped virus and xenotropic murine leukaemia virus (XMuLV) as the large enveloped virus (International Council for Harmonisation 2023a). Step yield will be measured at every design setting, on the unspiked control run that accompanies each clearance run, so that any operating region proposed for viral safety can also be checked against recovery.

Several things are outside this plan. The membrane itself is qualified by its manufacturer, and this study does not repeat that qualification; it characterizes the operating parameters the manufacturing site controls. The cumulative clearance claim for the drug substance is assembled in PCMR-001 from this step and from Step 6 and Step 8, and this plan covers only the contribution of Step 9. Product quality attributes are formed upstream and are addressed in the documents for the steps that form and polish them. The step does not by itself satisfy the cumulative clearance requirement for either model virus, and no part of this study should be read as claiming that it does.

2 Objectives

The study has five objectives.

- (i) Quantify the effect of Filtration volume (load) and Filtration pressure on MVM log reduction, on XMuLV log reduction and on step yield, across the characterization ranges in Table 6.1.
- (ii) Establish whether the two parameters interact, and whether either response is curved over the ranges studied, so that the operating region can be defined from a fitted model instead of from single-factor limits.
- (iii) Define the region of the two parameters over which the step meets its required log reduction for both model viruses.
- (iv) Derive a proven acceptable range for each parameter against each viral clearance attribute, and establish where the normal operating range sits in relation to it.
- (v) Classify each parameter under SOP-4001 and provide the ranges, controls and monitoring that the control strategy needs for Stage 2 (U.S. Food and Drug Administration 2011).

These objectives are answered by the study reported in PCR-009. Nothing in this plan is conditional on the outcome, and no objective may be dropped after the data are seen.

3 Related documents

Table 3.1 lists the documents this plan depends on and the documents that depend on it. PTP-001 frames the technology transfer that the whole campaign supports, RA-001 sets the scope of this study, PCMP-001 provides the framework common to every characterization plan in the campaign, and PCR-009 reports what is executed here.

Table 3.1: Related documents in the characterization package.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-009	Process Characterization Report (Small-Virus Retentive Filtration (Step 9))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Table 3.2 lists the controlled procedures and the validated analytical methods the study will use. All identifiers are placeholders for this synthetic document.

Table 3.2: Controlled procedures and validated methods for the study.

Refer- ence	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation

Refer- ence	Title	Type
AMV- 3017	XMuLV Infectivity Titre (TCID50)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The step is operated in dead-end mode at constant pressure. The anion exchange pool is passed through a pre-wetted small virus retentive filter, the filter is chased with buffer to recover the product held up in the device, and the filter is integrity tested after use (SOP-2012). The antibody passes through the retentive layer and virus particles are held within it. Retention is a size effect, so the margin is widest for a large enveloped virus and narrowest for a small non-enveloped one. MVM is the smallest of the model viruses used for a monoclonal antibody process, and it therefore sets the retention challenge for this step. XMuLV is much larger and is expected to be retained with a wide margin.

The step has been run on the Novacyte platform for three earlier antibodies (X-Mab, Y-Mab and Z-Mab) with the same filter family, and because the mechanism of removal is a physical one that does not depend on the product, the experience gained on those campaigns applies directly to A-Mab (CMC Biotech Working Group 2009). No difference in performance is expected here. Platform experience is the reason the study is small, and it is the expectation the study is designed to confirm and bound. It is not a substitute for the study, because clearance may be credited only from spiking data generated on the A-Mab process intermediate (International Council for Harmonisation 2023a).

Two mechanistic expectations shape the design. The first is that MVM log reduction falls slowly as the volumetric load rises, because retentive capacity is consumed as the membrane is loaded and the smallest particles are the first to find a path through. The second is that pressure acts on flux and on the compaction of the retentive layer, so any effect of pressure should be largest at high load, which makes it an interaction between the two parameters and not a main effect of either. Both expectations are testable in the design proposed in §6, and the study is arranged so that either can be refuted.

Step yield is expected to be high, and to be limited by the volume held up in the device and not by anything that happens to the product. The step is not expected to change any product quality attribute. Glycan, charge variant and aggregate attributes are formed in the production bioreactor and are polished at the chromatography steps, and filtration through a retentive membrane at moderate pressure has not induced aggregation or fragmentation on the platform (CMC Biotech Working Group 2009). For this reason the study measures viral clearance and yield, and carries no product quality design of its own.

4.2 Quality attributes in scope

Table 4.1 gives the two critical quality attributes this step contributes to. Both are viral safety attributes, both are of very high criticality, and both are cumulative process-level attributes: the acceptance criterion is a total log reduction across the whole purification train, not a value that any one step delivers. Criticality is assigned on the continuum used throughout the package, from the Tool #1 product of impact and uncertainty, with high and very high treated as critical (CMC Biotech Working Group 2009; International Council for Harmonisation 2023b).

Table 4.1: Critical quality attributes to which this step contributes.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6-20 log ₁₀ (cumulative)	VH	20
Viral clearance — XMuLV (enveloped)	viral safety	16.7-30 log ₁₀ (cumulative)	VH	20

The register names the step at which each attribute is set. MVM clearance is assigned to anion exchange and XMuLV clearance to low-pH inactivation, and this step contributes to both. The division of credit matters most for MVM. A low-pH hold acts on the lipid envelope, so it is credited with no clearance of a non-enveloped parvovirus, and the MVM requirement of 8.6 log₁₀ rests on the anion exchange step (PCR-008) and on this one, which is why the load limit set here is a viral safety control. XMuLV clearance is shared across all three viral steps and its cumulative requirement of 16.7 log₁₀ is correspondingly higher. The arithmetic that turns each cumulative requirement into a criterion for this step is given in §7, and the consolidated claim is made in PCMR-001.

4.3 Risk-based prioritization of parameters

RA-001 assessed every parameter of this step before the characterization campaign began, and its output is the study scope this plan executes. The assessment is summarized in Table 4.2.

Table 4.2: Pre-characterization risk assessment for the step (RA-001).

Parameter	Prospective failure mode	Effect	Sev.	Occ.	Det.	Initial RPN	Assigned study	Priority
Filtration volume (load)	Volumetric load above 105 L/m ²	Reduced MVM log-reduction (parvovirus breakthrough risk)	10	7	10	700	multivariate DoE	High

Parameter	Prospective failure mode	Effect	Sev.	Occ.	Det.	Initial RPN	Assigned study	Priority
Filtration pressure	Pressure outside filter operating limits	Potential impact on virus removal at high load	10	7	10	700	multivariate DoE	High

Both parameters carry the highest severity score available, because the attribute at risk is viral safety and a failure of retention would not be corrected anywhere downstream. Detection scores poorly at both parameters, because virus titre in the filtrate can be measured only in a deliberately spiked study and never on a manufacturing batch, so the process itself gives no direct in-process signal that retention has been lost. The integrity test detects a damaged device, which is a different failure. Both parameters were carried into the multivariate design by the risk ranking and filtering in RA-001, because the effect of each is expected to depend on the other, and both were ranked at high priority on severity.

RA-001 also fixed the ranges, and the reasoning is different for the two parameters. Filtration volume (load) is studied to 140 L/m², well above the upper edge of its normal operating range at 105 L/m², because the platform expectation is a slow decline in MVM clearance with load. A study confined to the normal range would confirm that the step works and would say nothing about where it stops working. Filtration pressure is studied over 8–30 psi, which brackets its normal operating range of 10–20 psi on both sides. The low edge is included because a long, slow filtration is a distinct operating state and not merely a milder version of the target, and the high edge because pressure is the parameter most likely to interact with load.

5 Materials and methods

5.1 Scale-down model and its qualification

The clearance claim rests entirely on a scale-down model, because virus cannot be introduced into a manufacturing batch. Viral safety guidance ICH Q5A requires the small-scale system to be representative of the commercial step before clearance is credited to it (International Council for Harmonisation 2023a), and SOP-1001 defines how that is demonstrated for this package. The model is a small-area device of the same membrane chemistry, the same retentive structure and the same housing geometry as the commercial filter, operated at the same volumetric load per unit area and the same pressure as the commercial step.

Qualification will compare the small-scale system against commercial-scale data on three points: the flux profile through the load, product recovery, and the quality of the filtrate. The comparison is made at the target operating point, with all parameters at their set-points, since the purpose is to establish that the two scales behave alike when the process is run as intended. Qualification is a prerequisite for the characterization runs. No DoE run will be executed until the scale-down model report has been approved, and a model that fails any of the three comparisons will be modified and requalified before the study starts.

The warrant the model provides is specific. It supports claims about how the retention of this membrane responds to volumetric load and pressure, because those are the variables the model reproduces, and it does not support claims about the pooling, hold and transfer operations that surround the step at commercial scale and are controlled by batch record. Both bounds are carried through to the conclusions of PCR-009.

Replicated centre points are built into both designs, 3 in the screening design and 4 in the response-surface design. The variation among them is the pure-error term against which lack of fit is judged (§5.5), and it is also the direct measure of how reproducibly the scale-down model performs at one setting.

5.2 Operation

The load material is the anion exchange pool. Its attributes are controlled to the in-process criteria for that step (PCP-008) and are not varied in this study, so they are identity- and quality-controlled inputs rather than characterized ones. The pool is filtered to the target volumetric load at constant pressure, chased with the same buffer to recover held-up product, and the device is integrity tested after use (SOP-2012).

For the clearance runs the load is spiked with a virus stock and then passed through a pre-filter before the retentive filter. Pre-filtration removes virus aggregates, which would otherwise be retained for reasons of size that have nothing to do with the performance of the membrane and would inflate the log reduction. Every other condition is held at platform values, so that the only deliberate differences between runs are the factor settings listed in Table 14.1 and Table 14.3.

5.3 Analytical methods

Two validated infectivity assays carry the study. MVM titre is measured by TCID50 with qPCR confirmation (AMV-3018) and XMuLV titre by TCID50 (AMV-3017). The log reduction factor for each run is the difference between the titre of the spiked load and the titre of the pooled filtrate, corrected for volume. Table 5.1 gives the validated performance of both methods.

Table 5.1: Validated performance of the analytical methods used for the study.

Method Title		Intermediate precision	LOQ	Accuracy (%)	Variance share
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	0.32 log10 SD	1.4 log10 TCID50/mL	96	0.3
AMV-3017	XMuLV Infectivity Titre (TCID50)	0.3 log10 SD	1.5 log10 TCID50/mL	96	0.28

Both assays report a titre on a log scale, so their intermediate precision is stated as a $\log_{10}$ standard deviation and not as a relative standard deviation. The MVM assay carries an intermediate precision of 0.32 $\log_{10}$ SD and the XMuLV assay 0.3 $\log_{10}$ SD. Assay variation propagates into every log reduction factor, so the standard error of an effect follows from it together with the replicated centre points of the design in which the effect is estimated (§5.5). An effect of the order of the assay precision will not be resolved by the screening design, and PCR-009 will report every effect with its standard error and its p-value and not as a bare magnitude.

The limit of quantitation sets a second and less obvious bound. When no virus is detected in the filtrate the log reduction cannot be reported as a number, and it is instead reported as at least the difference between the load titre and the limit of quantitation of the assay, which for the MVM method is a filtrate titre of 1.4 $\log_{10}$ TCID50/mL. Such a result is censored, not measured, and a censored value cannot be treated as an observation in a regression without biasing the fit. Runs that give a censored result will be identified in PCR-009 and the model for that response will be fitted to the runs that are not censored, with the censored runs reported alongside.

Step yield is not a virological measurement. It is calculated by mass balance across the step, from the protein mass in the load and the protein mass in the pooled filtrate, using the concentration method that supports the formulation step (AMV-3019, listed

in PCP-010). The full validation data for all three methods are held in their validation reports and are not reproduced here.

5.4 Sampling plan

Each run generates four kinds of sample. The load is sampled before spiking, to establish the starting material, and after spiking, to establish the input titre. The filtrate is collected in sequence as successive fractions across the load, so that clearance can be followed as a function of the volume already filtered. The fractions are pooled to give the cumulative filtrate, which is the sample the log reduction factor is calculated from. The post-use integrity test result is recorded for every run.

Fractionation is included so that clearance can be followed through the load. Break-through is a phenomenon of the later part of the load, and a single pooled sample averages the last fraction together with the first and hides it. The fractions cost assay capacity and nothing else, and they are what allows the study to locate where in the load a decline in clearance begins.

Centre-point runs are replicated: 3 replicates in the screening design and 4 in the response-surface design. Two controls accompany every clearance run. An unspiked run establishes yield and integrity without virus present, and a spike hold control establishes that the virus stock is stable over the duration of the run, so that a loss of titre is not mistaken for retention.

5.5 Statistical methods

The analysis follows SOP-1002. Both factors are analysed in coded units, with the low and high edges of the characterization range coded -1 and $+1$ and the centre of the design coded 0 . Coding puts the two parameters on one scale, so that effects can be compared directly and the fitted coefficients read as the change in the response over half the range studied.

The screening design is analysed by ordinary least squares with both main effects and their interaction. An effect is reported as twice the coded coefficient, which is the change in the response from the low edge of a factor to the high edge. The response-surface data are analysed with the full quadratic model, which adds a squared term for each factor to the main effects and the interaction. Significance is judged at $\alpha = 0.05$ throughout.

Model adequacy will be assessed on four things, and all four are reported in PCR-009: the coefficient of determination and its adjusted form, the predicted coefficient of determination from the PRESS statistic, the overall model F test, and an analysis of variance that partitions the residual into lack of fit and pure error. Pure error comes from the replicated centre points, which is why they are in both designs. A model with significant lack of fit describes the data it was fitted to but does not predict, and a model that fails this test will not be used to define an operating region.

The screening design and the response-surface design do different jobs, and this plan keeps them separate. The screening design identifies which factors are active and whether they interact. The response-surface model is the predictive model, and it alone is the basis of the design space and of the proven acceptable ranges in §8. Effect estimates from the screening runs will be reported as evidence of which factors matter, and will not be presented as predictions.

A factor with no significant effect on a response is a result and will be reported as one. It is evidence that the response is robust to that factor over the range studied, it is retained in the knowledge space on that basis, and it is classified accordingly under SOP-4001.

Commercial-scale capability will be estimated by Monte-Carlo simulation of 2,000 batches of the whole purification train, with every parameter varying about its set-point across its normal operating range. It is reported for each viral clearance attribute as a one-sided capability index against the cumulative drug substance requirement, together with the margin between the simulated mean and that requirement. The index is a property of the train and not of this step alone, and the contribution this step makes to it is the log reduction reported against the criteria of §7. This plan sets no numeric acceptance threshold for the capability index, because none is established for the campaign; capability is reported as a measure of margin and is interpreted against the criticality of the attribute (CMC Biotech Working Group 2009).

6 Study design

6.1 Factors, ranges and study type

Table 6.1 gives the parameters, their set-points, the ranges to be studied and the normal operating ranges. Table 6.2 gives the letter codes used for the factors in the design matrices and in the effect tables of PCR-009.

Table 6.1: Parameters, ranges and study type for the planned characterization.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Filtration volume (load)	L/m2	90	50-140	50-105	multivariate
Filtration pressure	psi	13	8-30	10-20	multivariate

Table 6.2: Factor legend for the design matrices.

Code	Factor	Unit	Studied in
A	Filtration volume (load)	L/m2	screening + RSM
B	Filtration pressure	psi	screening + RSM

Both parameters are assigned to the multivariate design, and each characterization range contains its normal operating range and extends well above it. Filtration volume (load) runs from 50-140 L/m2 against a normal range of 50-105 L/m2, so the two share their lower edge at 50 L/m2 and the studied range extends above the normal one to 140 L/m2. Filtration pressure runs from 8-30 psi against a normal range of 10-20 psi, which it brackets on both sides. The studied range is the knowledge space for this step. The design space will be a region inside it, and the normal operating range will lie within that region, touching the studied range only at the lower load edge.

The centre of the design is not the set-point of the process. It sits at 95 L/m2 and 19 psi, above the set-points of 90 L/m2 and 13 psi, because coding places the centre at the midpoint of each characterization range and both ranges extend further above the target than below it. Both centre settings lie inside the normal operating ranges of Table 6.1, so the centre point is a condition the process is licensed to run at and not a worst case. The worst case for this step is defined in §7, and it is a corner of the design and not its centre.

6.2 Screening design

The screening study is a full factorial in the 2 factors at two levels, augmented with 3 replicated centre points, giving 7 runs (Table 14.1). The design is not saturated. It estimates both main effects and their interaction independently, and the replicated centre points add an estimate of pure error and a test for curvature.

The purpose of this stage is to identify which of the two parameters drives each response, and whether the interaction expected in §4.1 is present. The screening fit will not be used to predict a response at an untested setting. Where curvature is detected at the centre points, the two-level design cannot describe it, and the response-surface stage exists for that reason.

6.3 Response-surface design

The response-surface study is a face-centred central composite design in the same 2 factors, with 19 runs across the two stages and 12 runs in this one (Table 14.3). It comprises the 4 factorial corners, 4 axial runs placed at the face centres of the design region, and 4 replicated centre points. The axial runs sit on the faces and not outside them, so that no run falls outside the characterization ranges given in Table 6.1.

This design supports the full quadratic model in both factors. It is the model from which the operating region and the proven acceptable ranges are derived, and it is the only model in this plan that is used to predict a response of this step. Its predictions are bounded by the ranges in Table 6.1, by the qualified scale-down model described in §5.1, and by the load material specification of the anion exchange step.

6.4 Univariate assessment

No parameter of this step is assigned to a univariate study. RA-001 assigned both parameters to the multivariate design (Table 4.2), and with only 2 factors in the design there is no case for separating them. The interaction is one of the things the study exists to test.

Three further inputs are known to affect virus filtration in general and are nonetheless not studied here. Buffer pH, conductivity and protein concentration have been shown across the platform to have little or no effect on the retention of either model virus over the ranges the process visits (CMC Biotech Working Group 2009), and the anion exchange step delivers a pool whose composition is controlled to a narrow specification (PCP-008). These inputs are therefore carried as controlled conditions and not as factors, and the justification for that is prior knowledge and not a study performed here.

The post-use filter integrity test is a procedural control and not a parameter. It is executed on every run of this study and on every commercial batch, and its result is

recorded as an attribute of the run. A failed integrity test invalidates the clearance result of that run.

7 Acceptance and decision criteria

This section states the criteria before the data exist. Table 7.1 gives what each measured response must achieve, and the basis of each criterion.

Table 7.1: Acceptance criteria for the measured responses.

Re- sponse	Quality attribute	Accep- tance criterion	Unit	Basis
MVM	Viral clearance	≥ 3.89	$\log_{10}$	Cumulative requirement less
LRF	— MVM		(this	the clearance credited to the
($\log_{10}$)	(parvovirus)		step)	other steps
XMuLV	Viral clearance	≥ 3.65	$\log_{10}$	Cumulative requirement less
LRF	— XMuLV		(this	the clearance credited to the
($\log_{10}$)	(enveloped)		step)	other steps

The two viral clearance criteria are step contributions and are derived, not specified. The drug substance requirement is cumulative, and the requirement placed on this step is that requirement less the clearance credited to the other steps of the train. On that basis this step must deliver at least $3.89 \log_{10}$ of MVM clearance and at least $3.65 \log_{10}$ of XMuLV clearance. Neither figure is a specification in its own right. Both move if the clearance credited to Step 6 or Step 8 changes, and PCR-009 will state the values in force when the study reports. Step yield has no criterion of this kind. It is a process performance attribute, it is reported for every run, and it is used to check that an operating region proposed on viral safety grounds is also a region the process can be run in.

The study is judged on four decisions, each of them made against evidence stated in PCR-009.

- (i) The scale-down model is qualified. Flux profile, recovery and filtrate quality at the target operating point are comparable with commercial-scale data under SOP-1001. This decision is made before the characterization runs, and a failure stops the study.
- (ii) Each response-surface model is adequate. The overall model is significant at $\alpha = 0.05$, lack of fit is not significant against pure error, and the predicted coefficient of determination supports use of the model outside the runs it was fitted to. A response whose model fails these tests is reported from the observed data only, and no region is claimed from it.

- (iii) An operating region is defined. The fitted model predicts at least $3.89 \log_{10}$ for MVM and at least $3.65 \log_{10}$ for XMuLV everywhere in the proposed region, including at its worst-case corner. For this step the worst-case corner is the high edge of Filtration volume (load) taken together with the pressure edge that the fitted model shows to be less favourable, and the region is not accepted on the strength of its centre.
- (iv) Each parameter is classified. Classification follows SOP-4001 from the demonstrated effect on a critical quality attribute and from the risk of leaving the design space in routine operation. A parameter with a demonstrated effect that is nonetheless easy to control and far from the region boundary is a well-controlled critical process parameter; one that is hard to control, or whose boundary is close to the normal operating range, is a critical process parameter.

These criteria apply within the characterization ranges of Table 6.1 and to material produced on the qualified scale-down model. A result outside those ranges is outside this plan. Any deviation from the plan will be recorded, investigated and impact assessed before the affected data are used, and the disposition will be reported in PCR-009.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range is a range of one parameter over which a quality attribute stays within acceptance, supported by data (International Council for Harmonisation 2009). The design space this study will define is multivariate, and the proven acceptable ranges complement it by stating, parameter by parameter, how far each one may move on its own. Both are reported in PCR-009. This section states how the ranges will be derived.

The acceptance basis is the one given in §7. For each viral clearance attribute the criterion is the step contribution back-calculated from the cumulative requirement, which is $3.89 \log_{10}$ for MVM and $3.65 \log_{10}$ for XMuLV. Step yield carries no such criterion and no proven acceptable range is derived for it.

Two analyses will be run for each combination of parameter and attribute. The first holds the other parameter at its set-point and evaluates the fitted response-surface model across the characterization range of the parameter of interest. The second propagates the normal operating range of the other parameter by Monte-Carlo, drawing 2,000 values at each of 81 points across the characterization range and taking the two-sided 95% predictive interval of the fitted model at each point. The first analysis answers what the parameter does when the rest of the process is at target. The second answers what it does when the rest of the process is anywhere in its normal range, which is the question a manufacturing site actually faces.

The decision rule is the same for both analyses. The proven acceptable range is the contiguous interval of the parameter, containing its set-point, over which the attribute stays within acceptance under that analysis. Under the first analysis the criterion is applied to the predicted mean. Under the second it is applied to the predictive interval, so the second range is the more demanding of the two and is expected to be the narrower. Where the range covers the whole characterization range, PCR-009 will say so and will state that the parameter was not limiting over the region studied. Where it is narrower, PCR-009 will name the attribute that binds it and the margin remaining between the range and the normal operating range.

Each proven acceptable range will be reported with a plot of the parameter against the attribute, with the acceptance limit drawn, the acceptable region shaded green, and the set-point and normal operating range marked. The at set-point and the NOR-propagated analyses appear as the two panels of that plot.

These ranges and the design space answer different questions, and the difference matters when the results are read. A proven acceptable range is univariate, in that it moves one parameter and holds or averages the other. The design space is the

multivariate region in which both parameters may move together. When both proven acceptable ranges span their whole characterization ranges, the binding constraint on the operating region is the worst-case corner of the multivariate region and no univariate range at all, and PCR-009 will present it that way. All of these ranges are bounded by the characterization ranges of Table 6.1, by the fitted model, and by the scale-down model on which the data were generated.

9 Data management and integrity

All raw data, calculations and reported results are governed by the ALCOA+ principles. Assay data are captured in the validated laboratory systems of the testing laboratory, process data are recorded on the executed run records for each scale-down run, and both are attributable to the analyst or operator who generated them.

The statistical analysis is performed with the seeded, version-controlled analysis code described in SOP-1002, so the tables and figures of PCR-009 can be regenerated from the raw data without manual steps. A second qualified person reviews the analysis against the raw data before the report is issued.

Changes to data after entry are made as tracked corrections with a reason recorded. No result is excluded from the analysis without a documented technical justification, and any exclusion is reported in PCR-009 together with its effect on the fitted model.

10 Roles and responsibilities

Table 10.1: Roles and responsibilities for the study.

Function	Responsibility
Process Development / MSAT	Owens this plan, executes the scale-down runs, performs the analysis and authors PCR-009
Virology testing laboratory	Prepares and titrates the virus stocks, performs the infectivity assays under AMV-3017 and AMV-3018
Analytical Development	Maintains the analytical methods and reviews the analytical sections of the report
Manufacturing Science	Confirms that the proposed ranges and controls are executable at the receiving site
Statistics	Reviews the design, the fitted models and the capability estimate
Quality Assurance	Approves this plan and PCR-009, and dispositions any deviation from the plan

Every function that approves this plan is named in the approval record at the front of the document (Table 1). The virology testing laboratory runs the infectivity assays under AMV-3017 and AMV-3018 as a service to the study and is not an approver. Execution of the study does not begin until every approver has signed.

11 Deliverables and schedule

The deliverable of this plan is PCR-009, the characterization report for the step. It reports the executed design, the fitted models, the operating region, the proven acceptable ranges, the capability estimate, the parameter classifications and any deviations from this plan. PCR-009 rolls up into PCMR-001, which consolidates the clearance claim across the three viral steps.

The work is sequenced in four phases. The scale-down model is qualified and its report approved. The screening design is then executed and analysed. The response-surface design follows, and is executed on the same model with the same load material specification. Analysis, capability estimation and reporting complete the sequence. No calendar dates are committed in this synthetic document. The sequence is what the plan commits to.

12 Risks and assumptions

Four risks could affect the study, and each has a defined response.

The scale-down model may not represent the commercial step across the whole range studied, even if it does so at the target operating point where it is qualified. This is the weightiest of the four, because every conclusion of the study inherits it. It is managed by qualifying against commercial data at the target point, by holding the model construction identical to the commercial device, and by stating the limitation in the conclusions of PCR-009 rather than leaving it to be inferred.

The virus stock may contain aggregates, which are retained by size and inflate the log reduction. Pre-filtration of the spiked load is the standard control and is built into the method (§5.2).

Clearance may exceed what the assay can measure, particularly for XMuLV, which prior knowledge suggests will be retained with a wide margin. The result is then censored at the limit of quantitation and the model for that response is weaker than the model for MVM. This is a known and accepted outcome; it is handled as described in §5.3, and it does not affect the MVM analysis, which is the analysis that governs the load limit.

Filter-to-filter and lot-to-lot variation in the membrane is not a factor in this design. Devices from more than one lot will be used across the runs where supply allows, and any run-order or lot effect will be examined in the residuals of the fitted models.

Three assumptions are made and are recorded here so they can be challenged. Platform experience with the same filter family on three earlier antibodies transfers to A-Mab, because the mechanism of removal is a physical one and does not depend on the product. The anion exchange pool used in the study is representative of the commercial load. The clearance credited to Step 6 and Step 8 stands as reported in PCR-006 and PCR-008, since the criteria in §7 are derived from it.

13 Approvals

This plan is effective when approved by all functions listed in the approval record at the front of the document (Table [1](#)). Approval of the plan authorises execution of the study as described. Any change to the objectives, the factors, the ranges or the acceptance criteria requires a revision of this plan, approved by the same functions, before the affected runs are executed.

14 Appendices A and B — Planned design matrices

The design matrices below are planned, not executed. They give the settings at which the runs will be made, and they carry no response columns, because no data exist when this plan is approved.

14.1 Appendix A — Planned screening design matrix

The planned screening runs are given in coded units in Table 14.1 and in natural units in Table 14.2. Response columns are absent because no data exist. Run order will be randomized at execution and the executed order reported in PCR-009.

Table 14.1: Planned screening design matrix, coded units.

Run	Type	A	B
1	factorial	-1	-1
2	factorial	-1	1
3	factorial	1	-1
4	factorial	1	1
5	center	0	0
6	center	0	0
7	center	0	0

Table 14.2: Planned screening design matrix, natural units.

Run	Type	Filtration volume (load)	Filtration pressure
1	factorial	50	8
2	factorial	50	30
3	factorial	140	8
4	factorial	140	30
5	center	95	19
6	center	95	19
7	center	95	19

14.2 Appendix B — Planned response-surface design matrix

The planned response-surface runs are given in coded units in Table 14.3 and in natural units in Table 14.4. The axial runs are placed at the face centres, so every setting lies within the characterization ranges of Table 6.1.

Table 14.3: Planned response-surface design matrix, coded units.

Run	Type	A	B
1	factorial	-1	-1
2	factorial	-1	1
3	factorial	1	-1
4	factorial	1	1
5	axial	-1	0
6	axial	1	0
7	axial	0	-1
8	axial	0	1
9	center	0	0
10	center	0	0
11	center	0	0
12	center	0	0

Table 14.4: Planned response-surface design matrix, natural units.

Run	Type	Filtration volume (load)	Filtration pressure
1	factorial	50	8
2	factorial	50	30
3	factorial	140	8
4	factorial	140	30
5	axial	50	19
6	axial	140	19
7	axial	95	8
8	axial	95	30
9	center	95	19
10	center	95	19
11	center	95	19
12	center	95	19

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.

International Council for Harmonisation. 2023a. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.

International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.